

Answers to the exercise #1 on XML

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9 October 2006

1 Threesome Shuffle

This time it is about shuffling three stacks of cards in the following way (the top of the stacks is on the top of the figure):

$$\begin{array}{ccccc} a_1 & a_2 & a_3 & & a_3 \\ b_1 & b_2 & b_3 & \rightarrow \dots \rightarrow & a_1 \\ c_1 & c_2 & & & a_2 \\ & d_2 & & & b_3 \\ & & & & b_1 \\ & & & & b_2 \\ & & & & c_1 \\ & & & & c_2 \\ & & & & d_2 \end{array}$$

If the three stacks are empty, then the result is empty too.

The empty stack is noted as usual `EMPTY` and the stack S with item e on the top is noted `PUSH( $e$ ,  $S$ )`.

Question. Define $\text{TSS}(S_1, S_2, S_3)$ as rewriting to the shuffling.

Answer.

$$\text{TSS}(\text{EMPTY}, \text{EMPTY}, \text{EMPTY}) \rightarrow \text{EMPTY} \quad (1)$$

$$\text{TSS}(A, B, \text{PUSH}(c, C)) \rightarrow \text{PUSH}(c, \text{TSS}(B, C, A)) \quad (2)$$

$$\text{TSS}(A, B, \text{EMPTY}) \rightarrow \text{TSS}(B, \text{EMPTY}, A) \quad (3)$$

Question. You will get extra points if

- your function is efficient;
- you make another version which returns `ERROR` if the number of items in the three stacks is not the same.

Answer. The solution with error reporting in case of the number of items is not the same in the three stacks is

$$\begin{aligned}
& \text{TSS}'(\text{PUSH}(a, A), \text{PUSH}(b, B), \text{PUSH}(c, C), D) \rightarrow \\
& \text{TSS}'(A, B, C, \text{PUSH}(c, \text{PUSH}(a, \text{PUSH}(b, D)))) \\
& \quad \text{TSS}'(\text{EMPTY}, \text{EMPTY}, \text{EMPTY}, D) \rightarrow \text{REV}(D) \\
& \text{TSS}'(\text{PUSH}(a, A), \text{PUSH}(b, B), \text{EMPTY}, D) \rightarrow \text{ERROR} \\
& \text{TSS}'(\text{PUSH}(a, A), \text{EMPTY}, \text{PUSH}(c, C), D) \rightarrow \text{ERROR} \\
& \quad \text{TSS}'(\text{PUSH}(a, A), \text{EMPTY}, \text{EMPTY}, D) \rightarrow \text{ERROR} \\
& \text{TSS}'(\text{EMPTY}, \text{PUSH}(b, B), \text{PUSH}(c, C), D) \rightarrow \text{ERROR} \\
& \quad \text{TSS}'(\text{EMPTY}, \text{PUSH}(b, B), \text{EMPTY}, D) \rightarrow \text{ERROR} \\
& \quad \text{TSS}'(\text{EMPTY}, \text{EMPTY}, \text{PUSH}(c, C), D) \rightarrow \text{ERROR}
\end{aligned}$$

Here we define efficiency as the number of rewrite steps. Therefore, we can improve TSS by taking items three by three as in TSS', but without reporting any errors.

$$\begin{aligned}
& \text{TSS}(\text{PUSH}(a, A), \text{PUSH}(b, B), \text{PUSH}(c, C)) \rightarrow \\
& \text{PUSH}(c, \text{PUSH}(a, \text{PUSH}(b, \text{TSS}(A, B, C)))) \\
& \quad \text{TSS}(\text{EMPTY}, \text{EMPTY}, \text{EMPTY}) \rightarrow \text{EMPTY} \\
& \text{TSS}(\text{PUSH}(a, A), \text{PUSH}(b, B), \text{EMPTY}) \rightarrow \\
& \text{PUSH}(a, \text{PUSH}(b, \text{TSS}(A, B, \text{EMPTY}))) \\
& \text{TSS}(\text{PUSH}(a, A), \text{EMPTY}, \text{PUSH}(c, C)) \rightarrow \\
& \text{PUSH}(c, \text{PUSH}(a, \text{TSS}(A, \text{EMPTY}, C))) \\
& \quad \text{TSS}(\text{PUSH}(a, A), \text{EMPTY}, \text{EMPTY}) \rightarrow \text{PUSH}(a, A) \\
& \text{TSS}(\text{EMPTY}, \text{PUSH}(b, B), \text{PUSH}(c, C)) \rightarrow \\
& \text{PUSH}(c, \text{PUSH}(b, \text{TSS}(\text{EMPTY}, B, C))) \\
& \quad \text{TSS}(\text{EMPTY}, \text{PUSH}(b, B), \text{EMPTY}) \rightarrow \text{PUSH}(b, B) \\
& \quad \text{TSS}(\text{EMPTY}, \text{EMPTY}, \text{PUSH}(c, C)) \rightarrow \text{PUSH}(c, C)
\end{aligned}$$

The version with error handling is straightforward:

$$\begin{aligned} & \text{TSS}(\text{PUSH}(a, A), \text{PUSH}(b, B), \text{PUSH}(c, C)) \rightarrow \\ & \text{PUSH}(c, \text{PUSH}(a, \text{PUSH}(b, \text{TSS}(A, B, C)))) \\ & \quad \text{TSS}(\text{EMPTY}, \text{EMPTY}, \text{EMPTY}) \rightarrow \text{EMPTY} \\ & \quad \text{TSS}(\text{PUSH}(a, A), \text{PUSH}(b, B), \text{EMPTY}) \rightarrow \text{ERROR} \\ & \quad \text{TSS}(\text{PUSH}(a, A), \text{EMPTY}, \text{PUSH}(c, C)) \rightarrow \text{ERROR} \\ & \quad \text{TSS}(\text{PUSH}(a, A), \text{EMPTY}, \text{EMPTY}) \rightarrow \text{ERROR} \\ & \quad \text{TSS}(\text{EMPTY}, \text{PUSH}(b, B), \text{PUSH}(c, C)) \rightarrow \text{ERROR} \\ & \quad \text{TSS}(\text{EMPTY}, \text{PUSH}(b, B), \text{EMPTY}) \rightarrow \text{ERROR} \\ & \quad \text{TSS}(\text{EMPTY}, \text{EMPTY}, \text{PUSH}(c, C)) \rightarrow \text{ERROR} \end{aligned}$$